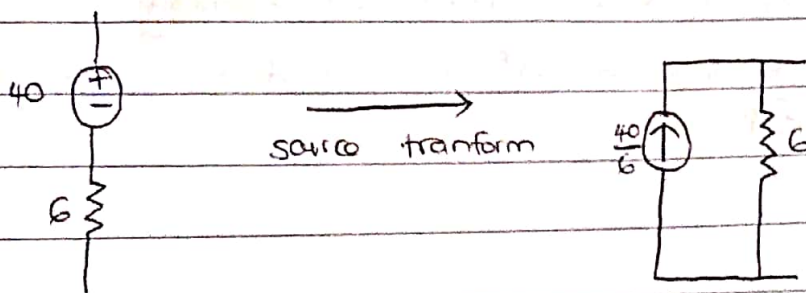


Source Transformation

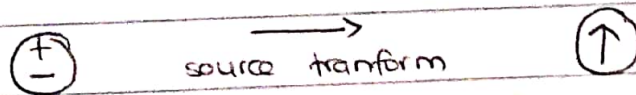
- > always change voltage $\oplus$ to Current $\uparrow$
- > If we have voltage and resistor we can source transform to get a $\uparrow$
- > ~~we are always going to source transform~~

> when **voltage source** is in **series** with a **resistor** we can source transform this $\oplus$ source ϕ $\parallel$ to obtain a **current source** and that $\uparrow$ source will be in **parallel** with a resistor.

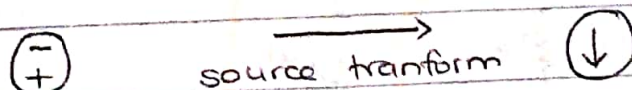


* the opposite is true !!

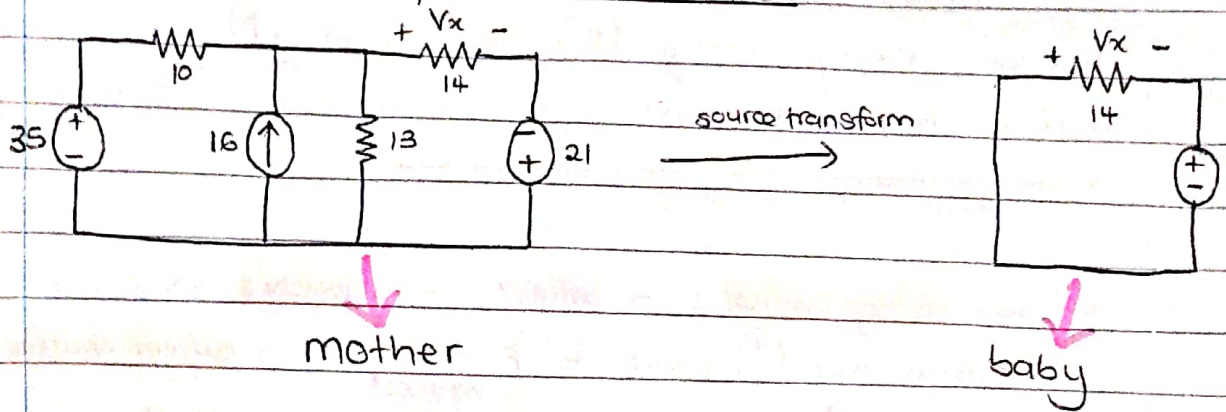
* that is we can source transform a current source in parallel with a resistor to obtain a voltage source in series with the resistor



* because of the polarity of the voltage source.



how ☺
 Example of what the questions will look like.

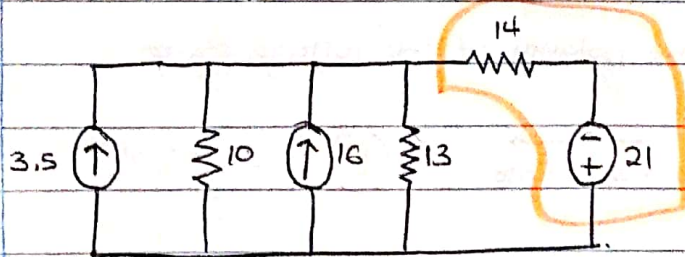
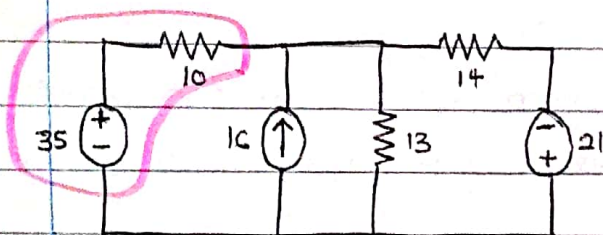


* **mother** has to be source transformed to **baby**

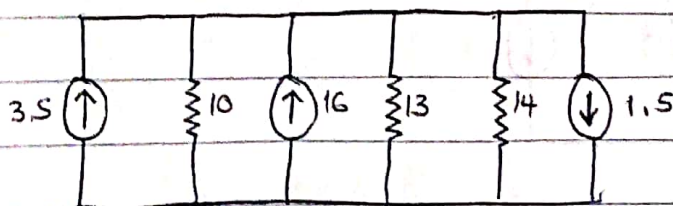
* this can be done in n^* , number of times until you get the required baby. ☺

Example 1

First step



Second step

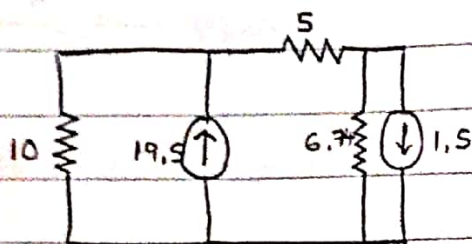
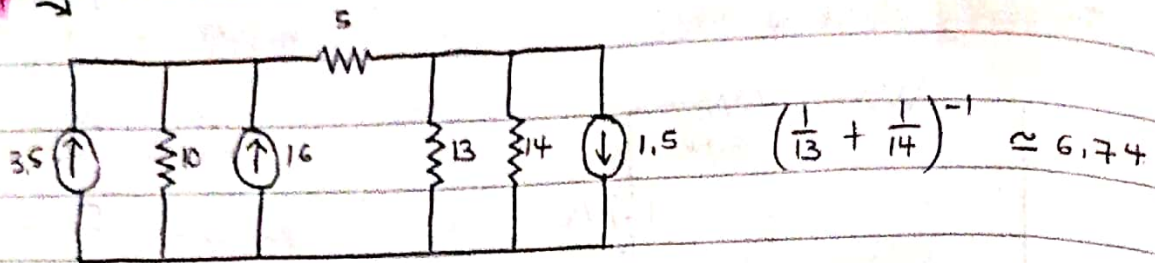


Note : all branches are sharing the same node !!

Resistance (Eq)

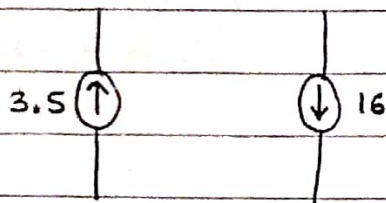
$$\left(\frac{1}{10} + \frac{1}{13} + \frac{1}{14}\right)^{-1} = 4.03 \text{ ohms}$$

modified circuit



* currents 3.5 and 16 all going up (same direction) hence take big value + smaller value and maintain the direction and shape of the bigger value!!! SSC

NB! Another scenario!



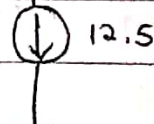
* different directions

* Bigger value - smaller value

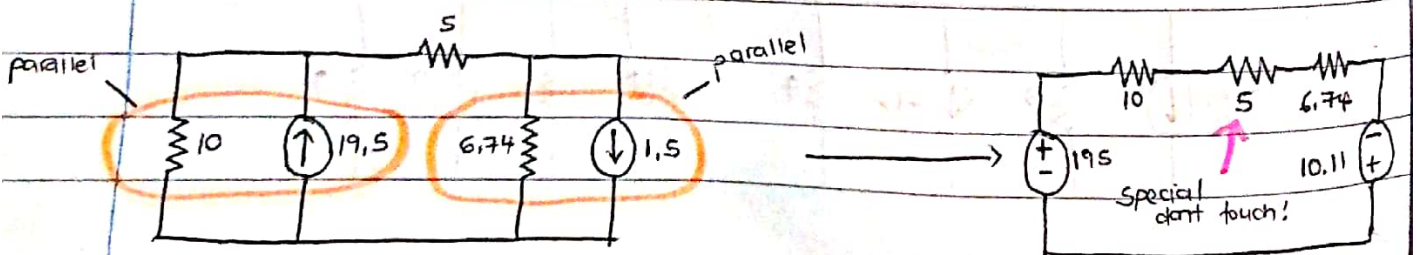
* take shape and direction of bigger value

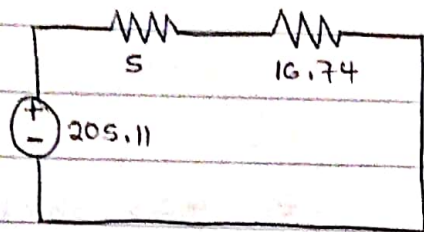
$$\therefore 16 - 3.5 = 12.5$$

* take direction of bigger value then

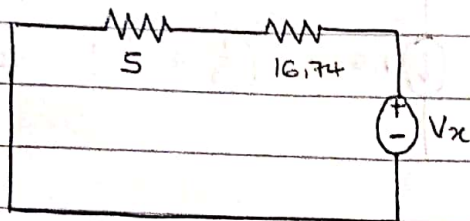


Back to example



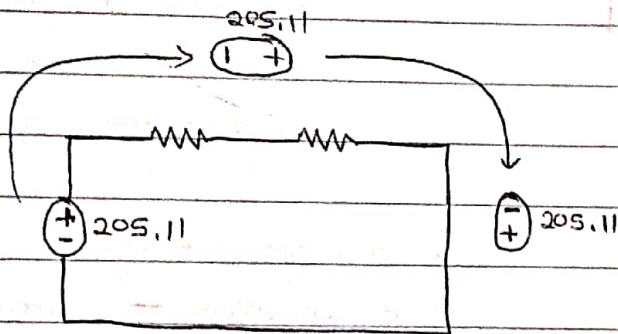


The **baby** would se like:



$$V_x = -205.11$$

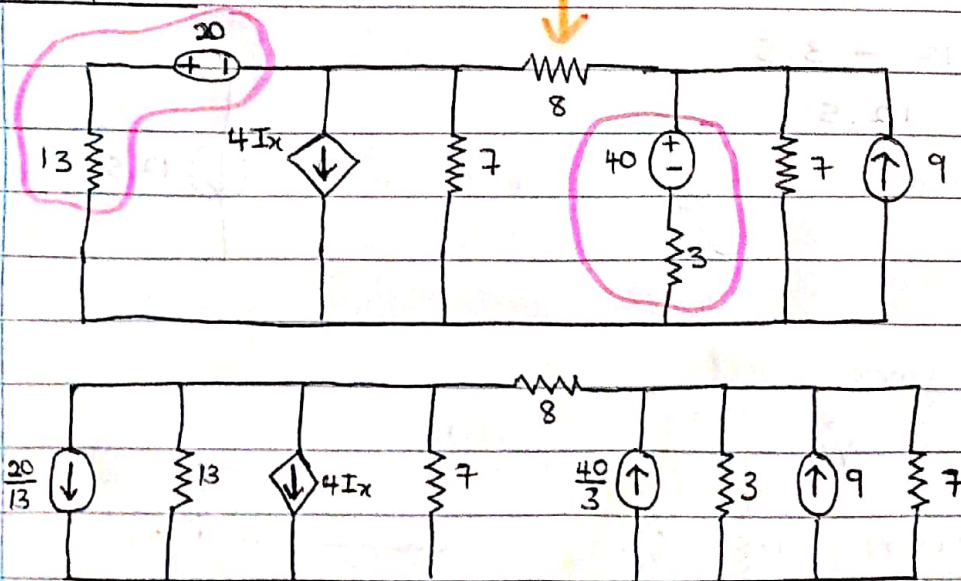
* **push** the voltage source to the position ~~th~~ as that on the **baby circuit**

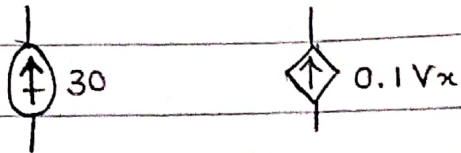
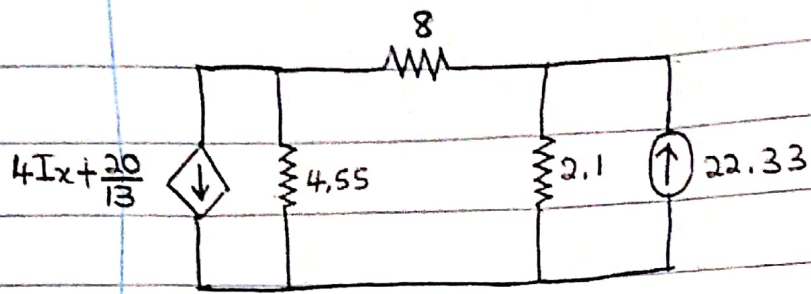


} different setting with
that of V_x
 $\therefore V_x = -205.11$

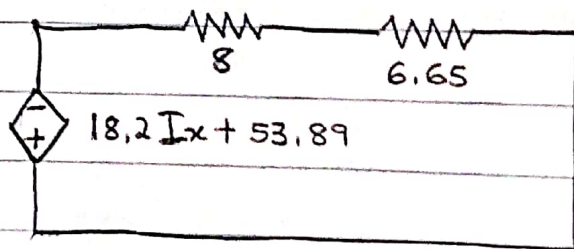
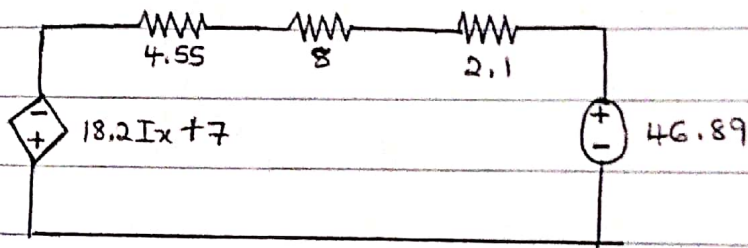
Example 2

special branch.





* the 30 will always be smaller i.e the dependent source will always be bigger. SSC !!



→ baby circuit